

F20DV Data Visualisation and Analytics

Individual Reflective Report - 2025/2026

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My contributions to this project covered the full pipeline, from the raw data through to the documentation that accompanied the finished application. Specifically, I built the data integration process, conducted the exploratory data analysis, structured the HTML page and wrote the shared CSS, implemented the cross-chart interaction logic in D3.js, and produced both the architecture guide and the user guide in the docs folder. Each of these areas came with its own set of challenges, and reflecting on them properly means being honest about what went wrong, not just what was completed.

Data Integration

The data pipeline required merging the Olympic history dataset with World Bank GDP data. The most significant challenge was a mismatch between identifier systems: Olympic records use NOC codes - DEN for Denmark, GER for Germany - while the World Bank uses ISO codes, DNK and DEU respectively. This was not immediately obvious, and my first instinct was to run the join directly. McKinney (2017) describes this class of failure as particularly dangerous because the operation completes without error while silently dropping all mismatched records. The impact would have been serious - a corrupted merged dataset with no visible warning, undermining every GDP-related finding produced from it. I caught it by running a test merge first, inspecting the unmatched rows carefully, and then building a manual mapping dictionary covering around thirty code pairs. Historical nations like the Soviet Union and East Germany needed separate handling, since they appear in the Olympic records but have no equivalent in modern GDP data; I flagged these as documented limitations rather than treating them as errors. A second issue was the GDP file arriving in wide format, with each year as its own column, which needed reshaping into long format before the join was even possible. Neither of these problems was in my original plan, and together they cost roughly a full day of work I had not budgeted for. What I learned is that auditing both datasets before writing any merge code - checking identifier schemas, file formats, and coverage gaps - is not preliminary work, it is where the real design decisions happen.

Exploratory Data Analysis

The EDA covered eight analytical areas, and the most valuable finding came from a linear regression on the GDP-medal relationship, which returned an R^2 of approximately 0.5. My initial interpretation of this was too strong - I read it as evidence that wealth drives Olympic success, which would have produced misleading claims in the data story. Cairo (2016) makes the point that a significant result should prompt further investigation rather than close it off, and that shift in thinking was the most important methodological correction I made during the project. Looking at the residuals changed the picture entirely: Cuba, Kenya and Jamaica consistently win more medals than their GDP would predict, while several high-income countries fall below the line. That over - and under-performer pattern became the most compelling thread running through the application, because it pushed the story away from a simple correlation and toward a more honest and interesting question about what GDP cannot explain. If I were approaching this analysis again, I would build the residual examination into the process from the start rather than arriving at it as a corrective step, since it consistently produced the most analytically useful output. I would also document analytical decisions in real time; Wilke (2019) notes that reproducibility depends not only on code but on the reasoning being recorded

alongside it, and on several occasions, I found myself uncertain why an earlier filtering choice had been made.

HTML Structure and CSS

I structured the HTML page and wrote the two shared stylesheets. The HTML work involved dividing the single-page application into clearly labelled sections - introduction, chart panels, and a narrative conclusion - so that other group members could attach their D3 components to specific container elements without overwriting each other's work. The CSS work involved defining shared colour variables, typography, and spacing rules that applied consistently across all views. Tufte (2001) argues that unnecessary visual variation adds cognitive load without adding information, and this was the principle behind centralising those decisions rather than leaving each chart to resolve them independently. The challenge that emerged mid-project was CSS specificity conflicts: D3-generated SVG elements were overriding page-level rules in unpredictable ways, and when the charts were first integrated together, several sections of the application looked visually inconsistent. The impact mattered because visual coherence is directly tied to how clearly the data story reads - a page that looks fragmented undermines trust in the content. I resolved this by introducing a selector naming convention across all chart scripts and coordinating with teammates to apply it consistently. This turned out to be as much a group communication problem as a technical one. The lesson is straightforward: front-end conventions need to be agreed and written down before development starts, not introduced retroactively when problems appear.

D3 Interactions

I implemented the bidirectional interaction between the choropleth map and the scatter plot, so that selecting a country on the map filters the scatter plot and selecting a data point on the scatter plot highlights the corresponding country on the map. Murray (2017) notes that interactions of this kind are among the most effective mechanisms for exploratory analysis, because they let users move between geographic and economic perspectives without losing their frame of reference. The technical challenge was preventing a recursive event loop: each chart's update was triggering the other, which without state management would cause an infinite update cycle and effectively crash the interaction. I resolved it by introducing a flag variable to track which chart had initiated a given update cycle, blocking the response from firing back. It was a small fix once understood but reaching that understanding took longer than expected and involved working through D3's event model more carefully than I had done before. This is an approach I would reuse directly in future projects - the pattern of using an initiator flag for bidirectional component communication is clean and reliable, and it would be my starting point for any similar interaction requirement.

Documentation, Group Work and Reflection

I wrote both documents in the docs folder. The architecture guide described the folder structure, the role of each JavaScript module, and how data flows through the application from load to render. The user guide was written for someone encountering the application for the first time, walking through each visualisation, explaining the controls, and suggesting a path through the data story. Both were written with the brief's requirement for maintenance-supporting documentation in mind. The main challenge was that I wrote the architecture guide after the application was largely finished. Some design decisions had evolved over the course of development without being recorded anywhere, which made parts of it harder to write accurately than they should have been. In future I would maintain it incrementally throughout the project, updating it as decisions are made rather than reconstructing them at the end.

On the group work side, the shared GitLab repository was not available for the first few weeks because it had not yet been set up. This meant the data pipeline and EDA work were done locally without any shared version control, which produced a compressed commit history that did not accurately reflect the development process. I worked around it by keeping local backups and pushing once the repository was ready, but it reinforced a habit I intend to keep going forward: initialise a local Git repository from the very first day of a project, regardless of whether a shared remote exists yet. Overall, the part of this project I am most satisfied with is the EDA, specifically the over- and under-performer analysis. The approach of checking residuals after a regression, rather than stopping at the coefficient, produced the most analytically valuable finding in the whole project and directly shaped what the visualisations were built to show. That is an approach I would apply as standard in any future data analysis work.

References

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